

# Quadratic Equations (Algebra)

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<https://leetcode.com/u/rajatkalia1/>  
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# Part I

## Theory

# Chapter 1

## Quadratic Equations

### 1.0.0.1 Linked Comprehension type Problems

**Comprehension 1:** Let  $a, b, c$  be three distinct real numbers and  $f(x)$  be a quadratic polynomial satisfying the equation

$$\begin{bmatrix} 9a^2 & 9a & 1 \\ 9b^2 & 9b & 1 \\ 9c^2 & 9c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(-2) \\ f(-3) \end{bmatrix} = \begin{bmatrix} 16a^2 + 10a \\ 16b^2 + 10b \\ 16c^2 + 10c \end{bmatrix}$$
. Let  $V$  be a point of local maxima of  $y = f(x)$  and  $A$  be the point where  $y = f(x)$  meets the x-axis and  $B$  be a point on  $y = f(x)$  such that  $AB$  subtends a right angle at  $V$

**Q1:** The curve  $f(x)$  is given by

- A)  $f(x) = -\frac{2}{9}x^2 + 2$
- B)  $f(x) = -\frac{x^2}{10} + 16$
- C)  $f(x) = -\frac{x^2}{16} + 10$
- D) None of these

**Q2:** The equation of chord  $AB$  can be

- A)  $5x + 6y + 15 = 0$
- B)  $6x + 5y + 15 = 0$
- C)  $65x + 27y + 106 = 0$
- D) None of these

**Q3:** The area of the region lying between the curve and the chord  $AB$  is

- A)  $\frac{3445}{8} \text{ unit}^2$
- B)  $\frac{3445}{64} \text{ unit}^2$
- C)  $\frac{3445}{128} \text{ unit}^2$
- D) None of these

Part II

Problem Set

## Chapter 2

# JEE Mains Problems

Q1: Let  $\alpha, \alpha + 2, \alpha \in \mathbb{Z}$ , be the roots of the quadratic equation

$x(x + 2) + (x + 1)(x + 3) + (x + 2)(x + 4) + \dots + (x + n + 1)(x + n - 1) = 4n$  for some  $n \in \mathbb{N}$ . Then  $n + \alpha$  is equal to :

[ JEE Mains 2nd April 2026 Shift 1 Q1 ]

A) 0

B) 1

C) 2

D) 3

{Solution: <https://www.youtube.com/watch?v=AVSUt9a0>}

Q2 : Let  $\alpha, \beta$  be the roots of the equation  $x^2 - 3x + r = 0$ , and  $\frac{\alpha}{2}, 2\beta$  be the roots of the equation  $x^2 + 3x + r = 0$ . If the roots of the equation  $x^2 + 6x = m$  are  $2\alpha + \beta + 2r$  and  $\alpha - 2\beta - \frac{r}{2}$ , then  $m$  is equal to :

[ JEE Mains 2nd April 2026 Shift 2 Q1 ]

A) -135

B) -567

C) 135

D) 567

{ Solution : <https://www.youtube.com/watch?v=4sKXoUOd8Is> }

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